Kanishk Pandey

+91 6367840193 | kanishkpandey1964@gmail.com | github.com/Kanishkp19 | linkedin.com/in/kanishk-pandey
23BDS027 | B.Tech in Data Science & AI | IIIT Dharwad | CGPA: 7.1 | SGPA Sem VI: 8.5

OBJECTIVE

B.Tech Data Science & AI student with hands-on experience building and fine-tuning Generative AI, deep learning, and multimodal systems from research literature. Passionate about implementing and optimising SOTA ML/DL/Foundation Models, with practical exposure spanning diffusion-based generation, self-supervised contrastive learning, ASR fine-tuning, and autonomous multi-agent pipelines. Seeking to contribute to model design, evaluation, and optimisation workflows in an applied research environment.

INTERNSHIP & EXPERIENCE

IIIT Dharwad – Clinical Speech & Language AI Aug 2025 – Present
Research Intern | Natural Language Processing / Speech AI | IIIT Dharwad

- Fine-tuned ASR (Automatic Speech Recognition) models on domain-specific medical corpora, significantly reducing Word Error Rate (WER) for complex pharmaceutical terminology; evaluated model outputs rigorously against annotated ground truth for accuracy and consistency.
- Designed a robust Speaker Diarization pipeline using clustering algorithms to segment overlapping doctor-patient dialogues, improving transcript accuracy for downstream NLP tasks.
- Engineered an abstractive Medical Summarization system using Large Language Models to convert raw transcripts into structured SOAP notes and clinical reports, ensuring consistent, high-quality structured text generation.

BrandContext.AI Apr 2025 – Jul 2025
Backend Developer Intern | Guide: Pulkit Khanna (Co-Founder & CTO) | Mumbai (Remote)

- Engineered scalable backend services integrating OpenAI, Grok, and Hugging Face APIs for production-grade AI workflows; containerised all services with Docker for consistent deployment.
- Designed and optimised computer vision pipelines using OpenCV and YOLO for real-time object detection and inference; managed async task queues with Celery and Redis for high-throughput workloads.
- Improved REST API latency and throughput via caching strategies and async processing; refactored monolithic codebases into modular, clean-code components.

PROJECTS

Multimodal Emotion Regulation Analysis | Self-Supervised Behavioral Modeling 2025

- Proposed a research-driven multimodal deep learning framework to infer emotional self-regulation using cross-modal incongruence signals and temporal dynamics, eliminating the need for discrete emotion labels.
- Applied self-supervised contrastive learning across face, audio, and text modalities; implemented a cross-modal incongruence encoder trained on aligned vs. shuffled multimodal samples.
- Modelled temporal stability using a Bi-LSTM over 10-30 timestep windows; introduced an interpretable Expressive Control Index (ECI) as a continuous behavioral metric.
- Validated cross-dataset generalisation: trained on IEMOCAP, evaluated on CMU-MOSEI with strict dataset separation.

Virtual Try-On (VTON) | Diffusion-Based Generative Clothing System 2026

- Implemented a SOTA diffusion-based virtual try-on system integrating Stable Diffusion Inpainting with IP-Adapter Plus for photo-realistic garment-conditioned image synthesis.
- Generated precise clothing masks via SCHP human parsing; preserved human pose and identity using ControlNet OpenPose conditioning.
- Optimised GPU inference with FP16 precision and attention slicing, reducing memory footprint and improving throughput for near real-time usage.

WikiSentinel | Autonomous Multi-Agent AI System 2025

- Built a real-time autonomous AI agent system to detect vandalism and misinformation in Wikipedia edits using multi-agent reasoning pipelines combining textual, visual, and policy-based signals.
- Implemented a scalable RAG pipeline over Wikipedia policy documents for explainable, policy-grounded decision-making; crafted and evaluated prompt-response pairs to improve agent reasoning quality.
- Integrated Llama 3 (Groq) and Gemini Vision for low-latency multimodal reasoning; achieved sub-second detection latency via event-driven stream processing.

COREP Reporting Assistant | Enterprise RAG & LLM Validation System 2025

- Built a RAG system over financial regulations using ChromaDB and dense sentence-transformer embeddings; engineered a validation layer with 11+ business rules ensuring CRR compliance and data consistency.
- Integrated structured LLM outputs with schema enforcement, retries, and fallback logic; generated comprehensive audit trails for regulatory traceability and explainability.

TECHNICAL SKILLS

ML / Deep Learning / NLP: PyTorch, TensorFlow, Hugging Face Transformers, RoBERTa, Sentence Transformers, BERTopic, FAISS, Contrastive Learning, Bi-LSTM

Generative AI & LLMs: Stable Diffusion, ControlNet, IP-Adapter, Diffusion Models, RAG Pipelines, Prompt Engineering, Structured Outputs, LangChain, Llama 3, Gemini, Foundation Models

Computer Vision: OpenCV, YOLO, MediaPipe
Speech & Language: ASR Fine-Tuning, Speaker Diarization, Medical Summarization, SOAP Note Generation, WER Optimisation
Backend & Systems: FastAPI, Flask, REST APIs, Redis, Celery, Docker, AWS, Async Processing
Databases: PostgreSQL, MongoDB, SQLite, ChromaDB, Pinecone DB, FAISS (Vector DBs)
Languages: Python (Expert), C, C++, JavaScript, HTML/CSS, LaTeX
Areas of Interest: Generative AI, Machine Learning, Deep Learning, Multimodal Learning, Computer Vision, NLP

POSITIONS OF RESPONSIBILITY

- Lead – Open Source Code Club, IIIT Dharwad
- Chess Lead and Captain – IIIT Dharwad Chess Team

EDUCATION

Indian Institute of Information Technology Dharwad 2023 – 2027
B.Tech in Data Science and Artificial Intelligence | CGPA: 7.1 | SGPA (Sem VI): 8.5

MDS Senior Secondary School, Udaipur 2023
CBSE – Class XII | 89%

MDS Senior Secondary School, Udaipur 2021
CBSE – Class X | 84%